

NOTES ON THE ORIGIN OF THE MANSFELD COPPER DEPOSITS.

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ABSTRACT.

A brief outline of the structure of the region is followed by observations on conditions underground at the Wolf shaft at Eisleben in June 1932, which indicate that the deposits are epigenetic.

THE copper deposits of the Mansfeld district are confined chiefly to a thin bed of black bituminous Permian shale which varies in thickness from five inches or less to twenty inches, the average being about nine inches. The shale underlies an area of 7000 square miles, but only a small fraction of it is known to be mineralized. Fifty square miles have been worked out, and known reserves are estimated at ten square miles.

The mines in this district have been worked for 744 years and have produced upwards of a million and a half tons of copper. The ore as mined contains from 2.5 to 2.8 per cent copper, and the annual production is about 25,000 tons of the metal. Silver, lead, and zinc are also produced.

The structure in the vicinity of Mansfeld is an open syncline with axis gently rising toward the west. The distance on the surface between the outcrops of the north limb and those of the south limb is 15 miles. The dip toward the axis varies from five to ten degrees. The formation is broken by many faults which are roughly parallel to the axis of the syncline. The fractures are pre-mineral, but there has been conspicuous post-mineral displacement.

My personal observations have been limited to the district around Eisleben and underground at the Wolf shaft. This shaft

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at the time of my visit (June 1932) was 557 meters deep. It is vertical, and at a considerable depth below the surface, it passes through the copper shale and continues well below into the underlying sandy conglomerate. From the station at the bottom of the shaft a horizontal drive provides a haulage way in the footwall to the ore bed—the copper shale.

The shale at this point is twelve inches thick and dips at six degrees from the horizontal. The pebbles in the underlying sandy conglomerate are nearly all white quartz as large as three inches in diameter but mostly smaller, and grade down in size toward the top to a coarse sand with grains from one-tenth to one-twentieth of an inch thick. There are also a few pebbles of black chert and jasper. The sandy beds immediately below the copper shale to a depth of from two to five feet are white—the Weissliegende, and below that they are stained faintly red with iron-oxide—the Rotliegende. The Weissliegende appears to be an integral part of the Rotliegende with a gradual reduction in size of grain toward the top and a rather abrupt loss of color that does not conform to planes of deposition or to cross-bedding, which is common.

The transition from white sands to the copper shale is abrupt, and in some places movement appears to have taken place on the contact. Going down the dip, the shale gradually thins down to an inch or less, and in some places where the bed is thin, copper mineralization extends down into the white sand a foot or so. The copper shale merges gradually upward into a dark impure limestone several feet thick. Where the overlying limestone has a red color the shale beneath it is barren of copper.

Farther down the dip the formations are cut off by a fault; a block several hundred feet wide has been dropped down about 60 feet; beyond the second fault the copper shale is found in its normal position. Crushed copper shale carrying ore has been dragged down along the faults. Near the faults are many small gash-like fissures a meter or so in length and a few centimeters wide filled with calcite and possibly some gypsum—a post-ore deposit. There are many faults in the district but mostly of small

throw, from two inches to two feet, rarely 150 to 300 feet. They are commonly short, parallel, and close together.

The chief ore minerals are bornite and chalcocite, the former predominating. Chalcopyrite and pyrite are also present, having been introduced later. The ore is not evenly distributed throughout the bed. It is more concentrated near the base and near the faults; away from the faults the bed is often much leaner. The ore minerals occur in minute particles irregularly disseminated throughout the shale or concentrated in streaks, generally parallel with the bedding. It also occurs in small nodules or concretions immediately under the shale in the sandstone and in the impure limestone above. Veins intersecting the shale and the underlying sandstone carry the nickel arsenides—niccolite, chloanthite, and maucherite with chalcopyrite, molybdenite, and uraninite. Zinc and lead sulphides occur in the upper part of the shale. The ores also carry small amounts of gold, silver, platinum, and vanadium.

Geologists from the early adolescence of the science have discussed the origin of the Mansfeld deposits. In 1756 Lehmann suggested that the copper was deposited in sea water with the shale and is, therefore, syngenetic. A majority of the geologists who have studied the deposits since have concurred in his opinion. Among these are B. von Cotta, von Groddeck, Steltzner, Bergeat, Lindgren, Pompeckj, Johannes Walther, Schneiderhöhn, R. Lang, R. H. Rastall, P. D. Trask,² and J. W. Gregory.³ The operating staff of the mine at the time of my visit also held this view.

In 1889 Beyschlag advanced the theory that the ores at Kamsdorf had their origin in the adjacent Permian eruptive rocks. He believed that the copper was leached from the eruptive and was carried in solution up the faults whence it spread through the shale where it was reduced by the bituminous matter. In 1893 Pošepny advocated this idea for Mansfeld, and it was adopted by Henry Louis, Beck, S. F. Emmons, J. E. Spurr, and Krusch and Vogt (1921) but later (1929) Krusch changed to the syngenetic theory.

² Trask, P. D.: *ECON. GEOL.*, 20: 746, 1925.

³ Gregory, J. W.: *Inst. Min. & Met.*, London, Dec., 1930.

It is not my purpose in these notes to weigh all the evidence for and against the two theories but merely to record my observations and their implications with respect to the origin of the deposits.

1. The concentration of ore near the base of the shale and near the faults suggests that the ore came from beneath the shale and entered it along the faults.

2. The leaching agent that removed the iron from the sand beneath the shale apparently was guided by the lower surface of the shale as it moved up the dip.

3. The Weissliegende were no doubt originally Rotliegende since the bleaching cuts indiscriminately through cross-bedding and other structures without reference to bedding planes.

4. The occurrence of red areas in the overlying calcareous beds where the shale is barren suggests that at those points the copper-bearing fluid found no opening by which it could enter the shale and pass through it to reduce the iron oxide in the overlying bed.

5. Ore in the sandstone beneath the shale where the latter is thin indicates that the ore is not syngenetic with the shale; neither could the ore in the cross fractures in the shale be syngenetic.

The chief stumbling block of the epigeneticists has been their inability to point directly to a magmatic source. But this is true of many copper deposits and there is much evidence that the relation between intrusives and ore deposits is more often tectonic than genetic. I am inclined to believe that the ore fluids at Mansfeld came from great depths through the east-west fractures into the underlying porous sandstone and moved up the dip under the bituminous shale, reducing the iron oxide in the sandstone and dissolving it; that the solution passed into the shale, especially where it was fractured, and deposited the sulphides in the strongly reducing environment. The east-west veins in the Harz mountains west of Mansfeld were probably contemporaneous and cogenetic. Bateman gives a similar explanation for the formation of hypogene chalcocite at the Bristol mine in Connecticut.⁴

There were apparently two periods of mineralization at Mansfeld; the first introduced bornite and chalcocite; and the second,

⁴ Bateman, Alan M., *ECON. GEOL.*, 18: 122-166, 1923.

arsenides of nickel and cobalt, chalcopyrite, pyrite, and other minor minerals. In its vast extent the copper shale of Mansfeld is not unlike the copper-bearing beds of Northern Rhodesia and of the Upper Peninsula of Michigan, but the porosity of the Rhodesian and Michigan formations permitted the movement of fluids through them, whereas at Mansfeld the porous bed lies immediately under the precipitating bed. That metallic copper was not precipitated in the Mansfeld sandstone as it was in the Michigan conglomerate may be due to the small amount of iron oxide in the Rotliegende which was overwhelmed by the strongly reducing hypogene fluids, whose temperature was probably too low to permit reaction with the silica.

Gregory, advocating a syngenetic origin, writes "The assortment of metals is not surprising in a sedimentary ore, but not the assembly of metals that would be expected to be derived from one eruptive magma." The evidence is clear that there were two periods of mineralization; the first, apparently, a normal hypogene fluid carrying bornite and chalcocite; and the second, similar to that which was responsible for the Sudbury (Canada) deposits. The second necessarily could not be syngenetic and the evidence apparently turns the scale in favor of an epigenetic origin for the first also.

For the privilege of making this study I am indebted to the officials and operating staff of the company at Eisleben who afforded me every facility, and who generously presented me with choice specimens from the copper shale, including the nickel arsenides, native silver, and a well preserved fossil fish *Palaeoniscus freieslebeni* Blainville.

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